

Report on designing and building a modular side-illuminated display device

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EE12001: Design Skills

University of Bath

Word count:

29 April 2025

Abstract

The project's goal was to use contemporary LED technology to design and construct a modular side-illuminated LED display that was modeled after the vintage Nixie tube. Using etched acrylic sheets and RGBY (Red, Green, Blue, Yellow) LEDs, the display simulates the vintage glow of Nixie tubes by illuminating numerals or symbols inscribed on each sheet. Important parts were a supporting MDF structure for alignment, unique PCBs for driving LEDs, and an Arduino Nano ESP32 for control. The project included soldering components onto PCBs, figuring out the right resistor values for various LED colors, and configuring the ESP32 to control LED sequences. The end result is a useful modular display that demonstrates the adaptability of embedded systems and contemporary production methods like laser cutting. This report improves knowledge of LED driver circuits, display technologies, and hardware-software interaction.

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Acronym Definition

LED: Light Emitting Diode

PCB: Printed Circuit Board

1.1 1. Introduction

1.1. Context

Older display technologies like Nixie tubes have been replaced by LED displays, which are now essential in contemporary embedded systems. The concept uses LED-illuminated laser-cut acrylic sheets to replicate Nixie looks. Modern clock designs, signage, and art installations all make use of these displays. The use of an Arduino Nano ESP32 allows software customization and hardware control to be seamlessly integrated.

1.2. Project Objectives

The main objectives of the project were:

- To design a modular LED-based display device.
- To integrate Red, Green, Blue, and Yellow LEDs for display purposes.
- To calculate and implement suitable resistor values for each LED colour.
- To develop a supporting MDF structure for aligning and securing the acrylic sheets.
- To program the ESP32 for controlling LED sequences.
- To gain practical experience with soldering and PCB assembly.

1.2 2. System Design

2.1. Hardware Design

- The system integrates several key components:
- **LEDs**: Red, Green, Blue, and Yellow (Farnell parts: 3619649, 3619644, 3619646, 3619650).
- **Resistors**: Appropriate resistor values were calculated based on LED forward voltage (V_{LED}) and current (I_{LED}). Using the following equation:
- $$R_B = \frac{V_{supply} - V_{LED}}{I_{LED}}$$
- With a 5V supply and typical LED forward voltages:
- Red: 2.0V
- Yellow: 2.1V
- Green: 3.0V
- Blue: 3.0V
- Assuming 20mA current:
- Red: RB=5−2.0/0.02=150Ω
- Yellow: RB=5−2.1/0.02=145Ω
- Green/Blue: RB=5−3.0/0.02=100Ω
- **PCB**: Custom-designed PCB to control the LEDs.
- **Acrylic Sheets**: Laser-cut with engraved symbols.
- **Arduino Nano ESP32**: Used for controlling the LED sequence.

2.2. Software Design

- The Arduino IDE was used to program the ESP32. The code initializes GPIO pins, defines LED sequences, and uses simple control algorithms to sequence the LEDs.
- **Main Code Features**:
- GPIO control for each LED.
- Delayed timing for sequencing LEDs.
- Serial monitor interface for debugging.

```
#define RED_PIN 2

#define GREEN_PIN 4

#define BLUE_PIN 5

#define YELLOW_PIN 18

void setup() {

 pinMode(RED_PIN, OUTPUT);

 pinMode(GREEN_PIN, OUTPUT);

 pinMode(BLUE_PIN, OUTPUT);

 pinMode(YELLOW_PIN, OUTPUT);

}

void loop() {

 digitalWrite(RED_PIN, HIGH);

 delay(500);

 digitalWrite(RED_PIN, LOW);

 digitalWrite(GREEN_PIN, HIGH);

 delay(500);

 digitalWrite(GREEN_PIN, LOW);

 digitalWrite(BLUE_PIN, HIGH);
```

```
  delay(500);
  digitalWrite(BLUE_PIN, LOW);
  digitalWrite(YELLOW_PIN, HIGH);
  delay(500);
  digitalWrite(YELLOW_PIN, LOW);
}
```

3. Results

- Hardware Assembly: Successful soldering of all components, including resistors and LEDs.
- Display Functionality: All LEDs illuminated correctly on the etched acrylic sheets.
- System Testing: LEDs were tested in sequence, ensuring consistent brightness and smooth transitions between colours.

4. Discussion

Strengths:

- The modular design allows for easy expansion and customization.
- The system works well for both static and dynamic display applications.

Challenges:

- Fine-tuning the resistor values for optimal current control.
- Ensuring uniform light distribution across the acrylic sheets.

Potential Improvements:

- Adding PWM control for brightness adjustment.
- Implementing more complex animation sequences using additional control logic.

5. In conclusion

The design and fabrication of a modular LED display that resembles Nixie tubes was successfully shown by this project. The combination of hardware and software, such as the Arduino ESP32 and custom PCBs, produced a useful and eye-catching display. Through this research, knowledge of circuit design, embedded systems, and contemporary production methods like laser cutting was improved.

6.References

1. Farnell. **LED Datasheets**. Retrieved from Farnell.
2. Arduino. **ESP32 Technical Documentation**. Retrieved from Arduino.

7.Appendices

Appendix A: Arduino Code

```
cpp
Copy
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#define GREEN_PIN 4
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#define YELLOW_PIN 18

void setup() {
  pinMode(RED_PIN, OUTPUT);
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  pinMode(YELLOW_PIN, OUTPUT);
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  delay(500);
  digitalWrite(RED_PIN, LOW);
  digitalWrite(GREEN_PIN, HIGH);
  delay(500);
  digitalWrite(GREEN_PIN, LOW);
  digitalWrite(BLUE_PIN, HIGH);
  delay(500);
  digitalWrite(BLUE_PIN, LOW);
  digitalWrite(YELLOW_PIN, HIGH);
  delay(500);
  digitalWrite(YELLOW_PIN, LOW);
}
```